Industrial Internship Report on

Expense Tracker

Prepared by

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Executive Summary
This report details my industrial internship through upskill Campus and The IoT Academy, in collaboration with UniConverge Technologies Pvt Ltd (UCT). The internship involved weekly learning of Core Java concepts and developing a small project – a Console-based Expense Tracker Application. This console Java app lets users add/view expenses, manage categories, and generate simple spending summaries. It is implemented using object-oriented design with separate classes (Expense , ExpenseManager , and Main). An ArrayList stores expense records, and the Scanner class handles user input. The menu-driven interface is implemented with a loop and switch statements to process user choices. The GitHub repository for the project is https://github.com/ttsmcks/upskillCampus.

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1 Preface

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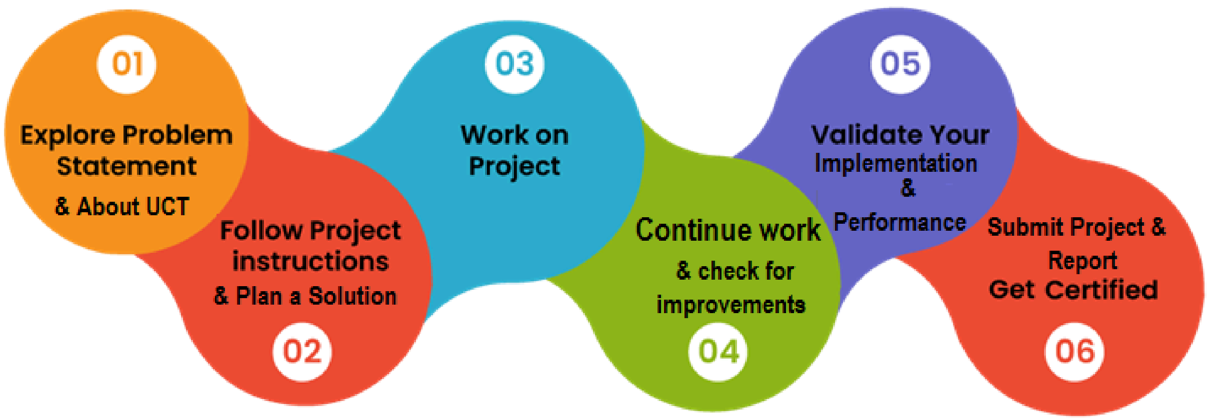


The selected project, Console-based Expense Tracker Application, was simple but useful for learning how to organize a Java program into modules and functions. This experience helped me understand the importance of planning, coding, testing, and improving a project step by step. I would like to thank the internship organizers, mentors, and everyone who supported me throughout this program

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

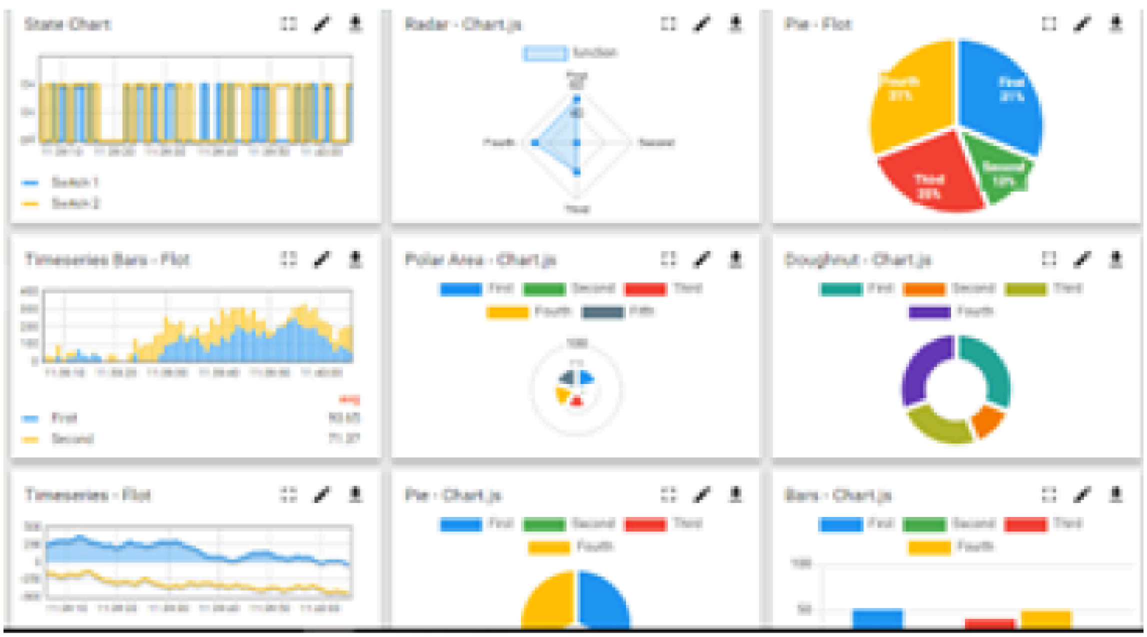
i. UCT IoT Platform (Insight)

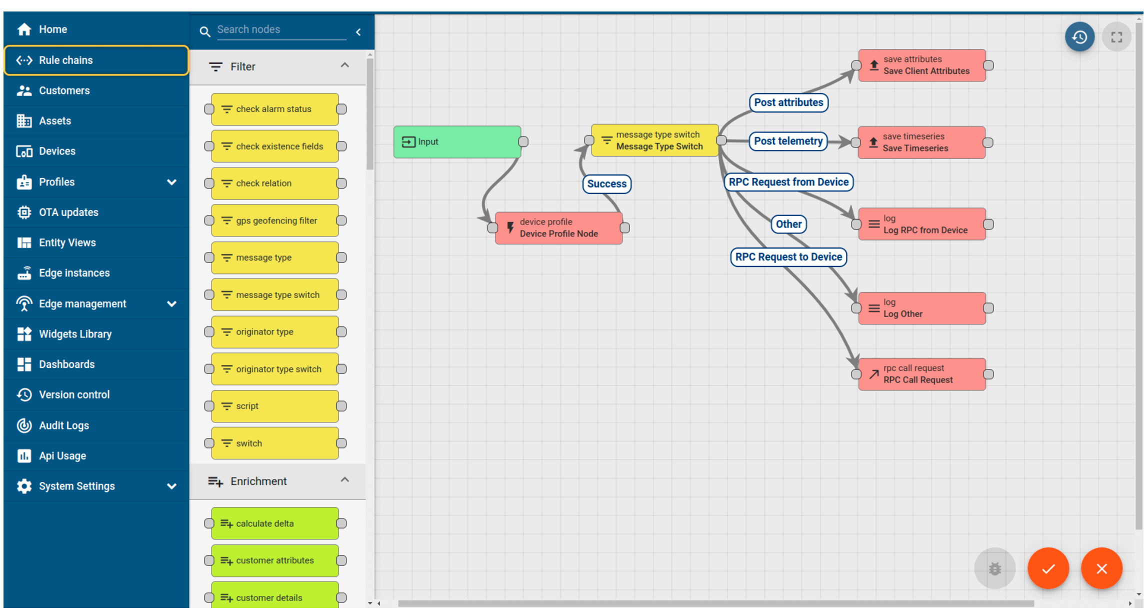
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





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ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

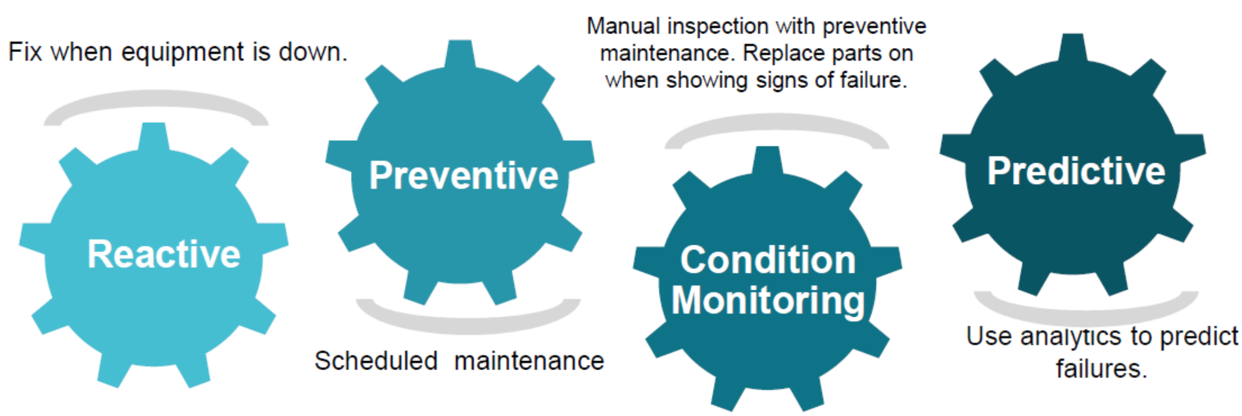
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

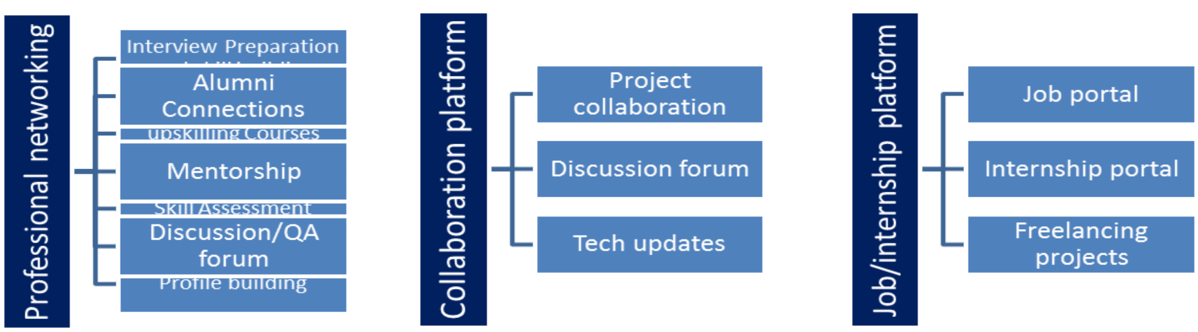
upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Java learning materials and notes

[2] Online Java tutorials and documentation

[3] Internship problem statement document

2.6 Glossary

Terms	Acronym
Java	Programming language used for this project.
OOP	Object-Oriented Programming.
IDE	Integrated Development Environment
Expense Tracker	Application used to store and manage expenses.

3 Problem Statement

The assigned project was to develop a Console-based Expense Tracker Application using Core Java. The application should allow the user to add expenses, view expense records, manage categories, filter data, and generate simple reports. The purpose of the project was to create a small but useful application that helps users track their personal spending in an easy way through a console interface.

4 Existing and Proposed solution

There are many expense tracking applications available, but most of them are mobile or web-based and have many advanced features that are not necessary for a beginner project.

The proposed solution for this internship is a simple console-based Java application that focuses on the main features only. It will allow users to:

- Add new expenses.
- View recorded expenses.
- Manage categories.
- See total spending.
- Generate basic summaries.

4.1 Code submission (Github link)

GitHub repository: https://github.com/ttsmcks/upskillCampus

4.2 Report submission (Github link)

GitHub repository:
https://github.com/ttsmcks/upskillCampus/ExpenseTracker_Arvind_Gotike_USC_UCT.pdf

5 Proposed Design/ Model

The project follows a simple flow: 1. The user opens the console application. 2. The system shows a menu with options (Add Expense, View Expenses, Total Expense, Exit). 3. The user chooses an option and provides input (using Scanner). 4. The entered data is processed and stored in memory (an ArrayList of expenses). 5. The system displays the requested output or calculates totals.

6 Performance Test

Since this is a small Core Java project, the main performance focus is on correctness and smooth functioning of the console menu. The important test points were:

- Add a valid expense and verify it is stored correctly.
- View all expenses after adding them.
- Add multiple expenses and check if total calculation is correct.
- Try entering an invalid amount and observe how the application handles it.
- Check that menu navigation works without errors.

6.1 Test Plan/ Test Cases

- Add a valid expense.
- View all expenses.
- Add multiple expenses and calculate total.
- Enter an invalid amount.
- Test menu navigation and invalid choices.

6.2 Test Procedure

The application was run in a Java environment, and each menu option was tested using sample inputs. The console output was observed to ensure that the functionality works as expected.

6.3 Performance Outcome

The application worked properly for the tested menu options. The expense details were stored and displayed correctly, and the total spending was calculated as expected.

7 My learnings

During this internship, I learned the basics of Core Java and understood how to build a small console application step by step. I also improved my understanding of classes, objects, methods, loops, conditions, and user input handling. This internship helped me gain confidence in writing Java programs and made me more comfortable with basic project development.

8 Future work scope

This project can be improved in the future by adding:

- File handling for permanent storage of expense data.
- Editing and deleting expense entries.
- Date-wise filtering and monthly reports.
- A simple login or user management system.

These additions would make the application more useful and closer to a real-world expense tracking system.